과제 6

1.

$$y^* := B^T y = B^T X\beta + B^T Z\gamma + B^T \epsilon$$

$$= B^T Z\gamma + B^T \epsilon$$

$$E y^* = 0 \ , \ Var(y^*) = Var(B^T y) = B^T V B$$

$$y^* \sim N(0, B^T V B)$$

$$f(y^*) = \frac{1}{(2\pi)^{p/2} |B^T V B|} e^{-\frac{1}{2} y^T B (B^T V B)^{-1} B^T y}$$

$$l_R = -\frac{p}{2}\log 2\pi - \frac{1}{2}\log|B^T V B| - \frac{1}{2} y^T B (B^T V B)^{-1} B^T y$$

$$\propto -\frac{1}{2}\log|B^T V B| - \frac{1}{2} y^T B (B^T V B)^{-1} B^T y$$

$$(y - X\hat{\beta})^T V^{-1} (y - X\hat{\beta})$$

$$= y^T V^{-1} y - y^T V^{-1} X\hat{\beta} - \hat{\beta}^T X^T V^{-1} y + \hat{\beta}^T X^T V^{-1} X\hat{\beta}$$

$$= y^T V^{-1} y - y^T V^{-1} X (X^T V^{-1} X)^{-} X^T V^{-1} y - y^T V^{-1} X (X^T V^{-1} X)^{-} X^T V^{-1} y$$

$$+ y^T V^{-1} \underbrace{X (X^T V^{-1} X)^{-} X^T V^{-1} X}_{=X} (X^T V^{-1} X)^{-} X^T V^{-1} y$$

$$= y^T V^{-1} y - y^T V^{-1} X (X^T V^{-1} X)^{-} X^T V^{-1} y$$

$$= y^T (V^{-1} - V^{-1} X (X^T V^{-1} X)^{-} X^T V^{-1}) y$$

$$= y^T B (B^T V B)^{-1} B^T y$$

Since X and B are fixed, we orthogonalize X and B s.t $X^T X = I$, $B^T B = I$, $X^T B = 0$

$$\Rightarrow (X \;\; B) \text{ is orthogonal}$$

$$\Rightarrow |V| = |(X \;\; B)^T V (X \;\; B)|$$

$$= \left| \begin{pmatrix} X^T V X & X^T V B \\ B^T V X & B^T V B \end{pmatrix} \right|$$

$$= |B^T V B| \, \underbrace{|X^T V X - X^T V B (B^T V B)^{-1} B^T V X|}_{A_{11.2}}$$

$$\{(X \;\; B)^T V (X \;\; B)\}^{-1} = (X \;\; B)^T V^{-1} (X \;\; B)$$

$$\text{upper-left} \quad X V^{-1} X$$

$$\Rightarrow \underbrace{\{X^T V X - X^T V B (B^T V B)^{-1} B^T V X\}}_{A_{11.2}^{-1}}{}^{-1} = X^T V^{-1} X$$

$$\Rightarrow |X^T V X - X^T V B (B^T V B)^{-1} B^T V X|^{-1} = |X^T V^{-1} X|^{-1}$$

$$\Rightarrow |V| = |B^T V B| \, |X^T V^{-1} X|^{-1}$$

$$\Rightarrow |B^T V B| = |V| \, |X^T V^{-1} X|$$

$$\Rightarrow \log |B^T V B| = \log |V| + \log |X^T V^{-1} X|$$

$$\Rightarrow l_R = -\frac{p}{2}\log 2\pi - \frac{1}{2}\log|B^T V B| - \frac{1}{2} y^T B (B^T V B)^{-1} B^T y$$

$$\propto -\frac{1}{2}\log|B^T V B| - \frac{1}{2} y^T B (B^T V B)^{-1} B^T y$$

$$= -\frac{1}{2}\log|V| - \frac{1}{2}\log|X^T V^{-1} X| - \frac{1}{2}(y - X\hat{\beta})^T V^{-1}(y - X\hat{\beta})$$

$\boxtimes$

$\boxed{2}$.

$$n_j := \sum_{i=1}^{n} \mathbb{1}\{x_i = j\}, \quad y_j := \sum_{i=1}^{n} y_i \mathbb{1}\{x_i = j\}, \quad j = 0, 1$$

$$p_0 := \mathbb{P}(Y=1 \mid X=0) = \frac{e^{\beta_0}}{1+e^{\beta_0}}$$

$$p_1 := \mathbb{P}(Y=1 \mid X=0) = \frac{e^{\beta_0+\beta_1}}{1+e^{\beta_0+\beta_1}}$$

$$\mathcal{L}(\beta_0, \beta_1) = p_0^{y_0}(1-p_0)^{n_0-y_0}\, p_1^{y_1}(1-p_1)^{n_1-y_1}$$

$$\ell(\beta_0, \beta_1) = y_0 \log p_0 + (n_0 - y_0)\log(1-p_0) + y_1 \log p_1 + (n_1 - y_1)\log(1-p_1)$$

For $j = 0, 1$

$$\frac{\partial \ell}{\partial p_j} = \frac{y_j}{p_j} - \frac{n_j - y_j}{1-p_j} = 0 \iff \hat{p}_j = \frac{y_j}{n_j}$$

$$\frac{\partial^2 \ell}{\partial p_j^2} = -\frac{y_j}{p_j^2} - \frac{n_j - y_j}{(1-p_j)^2} < 0 \quad \forall\, 0 < y_j < n_j.$$

$0 < y_0 < n_0,\ 0 < y_1 < n_1,$

$$\hat{\beta}_0 = \operatorname{logit}(\hat{p}_0) = \log \frac{y_0/n_0}{1 - y_0/n_0} = \log \frac{y_0}{n_0 - y_0}$$

$$\hat{\beta}_1 = \operatorname{logit}(\hat{p}_1) - \operatorname{logit}(\hat{p}_0)$$

$$= \log \frac{y_1}{n_1 - y_1} - \log \frac{y_0}{n_0 - y_0}$$

$$= \log \frac{y_1(n_0 - y_0)}{(n_1 - y_1)y_0}$$

If $y_j = 0$ or $y_j = n_j$ for some j,

the finite MLE does not exist.

$$\therefore \quad \hat{\beta}_0 = \log \frac{y_0}{n_0 - y_0}, \quad 0 < y_0 < n_0, \; 0 < y_1 < n_1$$

$$\hat{\beta}_1 = \log \frac{y_1 (n_0 - y_0)}{(n_1 - y_1) y_0}$$

If $y_j = 0$ or $y_j = n_j$, MLE diverges $\blacksquare$

$\boxed{3}$.

$$k(Y_m \mid Y_o ; \theta) = \frac{f(Y_c ; \theta)}{g(Y_o ; \theta)}$$

$$\Rightarrow \log k(Y_m \mid Y_o ; \theta) = \log f(Y_c ; \theta) - \log g(Y_o ; \theta)$$

$$\begin{aligned}
S^*(Y_o ; \theta) &= \frac{\partial}{\partial \theta} \log g(Y_o ; \theta) \\
&= \frac{1}{g(Y_o ; \theta)} \frac{\partial}{\partial \theta} g(Y_o ; \theta) \\
&= \frac{1}{g(Y_o ; \theta)} \int \frac{\partial}{\partial \theta} f(Y_c ; \theta) \, dY_m \\
&= \int \underbrace{\frac{\partial}{\partial \theta} \log f(Y_c ; \theta)}_{s} \underbrace{\frac{f(Y_c ; \theta)}{g(Y_o ; \theta)}}_{k} dY_m \\
&= \mathbb{E}_\theta [S(Y_c ; \theta) \mid Y_o]
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \theta^T} \log k(Y_m \mid Y_o ; \theta) &= \frac{\partial}{\partial \theta^T} \log f(Y_c ; \theta) - \frac{\partial}{\partial \theta^T} \log g(Y_o ; \theta) \\
&= S(Y_c ; \theta)^T - S^*(Y_o ; \theta)^T
\end{aligned}$$

$$\frac{\partial}{\partial \theta^T} \log k = \frac{1}{k} \frac{\partial k}{\partial \theta^T}$$

$$\Rightarrow \frac{\partial}{\partial \theta^T} k(Y_m \mid Y_o ; \theta) = k(Y_m \mid Y_o ; \theta) \{ S(Y_c ; \theta) - S(Y_o ; \theta) \}^T$$

$$S^*(Y_o;\theta) = \int S(Y_c;\theta)\, k(Y_m \mid Y_o;\theta)\, dY_m$$

$$\frac{\partial}{\partial\theta^T} S^*(Y_o;\theta) = \int \frac{\partial}{\partial\theta^T} S(Y_c;\theta)\, k(Y_m \mid Y_o;\theta)\, dY_m$$

$$+ \int S(Y_c;\theta) \frac{\partial}{\partial\theta^T} k(Y_m \mid Y_o;\theta)\, dY_m$$

$$= -E_\theta[B(Y_c;\theta) \mid Y_o]$$

$$+ E_\theta[S(Y_c;\theta)[S(Y_c;\theta) - S^*(Y_o;\theta)]^T \mid Y_o]$$

$$= -E_\theta[B(Y_c;\theta) \mid Y_o] + E[S(Y_c;\theta)S(Y_c;\theta)^T \mid Y_o]$$

$$- S^*(Y_o;\theta)\, S^*(Y_o;\theta)^T$$

$$\Rightarrow -\frac{\partial^2}{\partial\theta\partial\theta^T} \log g(Y_o;\theta)$$

$$= E_\theta[B(Y_c;\theta) \mid Y_o] - E_\theta[S(Y_c;\theta)S(Y_c;\theta)^T \mid Y_o]$$

$$+ S^*(Y_o;\theta)\, S^*(Y_o;\theta)^T$$

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